Practical No 02

a) Develop an Android application implementing Explicit and Implicit Intents.

ProjectName: IntentDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">
   <Button
      android:id="@+id/btnExplicit"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Explicit Intent"/>
   <Button
      android:id="@+id/btnImplicit"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Implicit Intent"
      android:layout_marginTop="20dp"/>
</LinearLayout>
```

Create Second Activity.

Right Click: java package → com.example.projectname -> New→ Activity → Empty Views Activity

Activity Name: SecondActivity
Click: Finish

Design Second Activity Layout.
Open: res→ layout → activity_second.xml

File Name: activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="24sp"/>
</LinearLayout>
```

File Name: MainActivity.java.

```
package com.example.intentdemo;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
```

```
    Button btnExplicit, btnImplicit;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicit = findViewById(R.id.btnImplicit);
        // Explicit Intent
        btnExplicit.setOnClickListener(v -> {
            Intent intent = new Intent(
                    MainActivity.this,
                    SecondActivity.class
            );

            startActivity(intent);
        });
        // Implicit Intent
        btnImplicit.setOnClickListener(v -> {
            Intent intent = new Intent(
                    Intent.ACTION_VIEW,
                    Uri.parse("https://www.google.com")
            );
            startActivity(intent);
        });
    }
}
```

b) Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).

Project Name: EventHandlingDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="20dp">

   <Button
      android:id="@+id/btnClick"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me"/>

   <Button
      android:id="@+id/btnLongClick"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Long Press Me"
      android:layout_marginTop="20dp"/>

</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.eventhandlingdemo;

import android.os.Bundle;

import android.widget.Button;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;



public class MainActivity extends AppCompatActivity {

   Button btnClick, btnLongClick;



   @Override

   protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnClick = findViewById(R.id.btnClick);
        btnLongClick = findViewById(R.id.btnLongClick);

        // OnClickListener
        btnClick.setOnClickListener(v -> {

            Toast.makeText(
                    MainActivity.this,
                    "Button Clicked",
                    Toast.LENGTH_SHORT
            ).show();
        });
        // OnLongClickListener
        btnLongClick.setOnLongClickListener(v -> {

            Toast.makeText(
                    MainActivity.this,
                    "Button Long Pressed",
                    Toast.LENGTH_SHORT
            ).show();
            return true;
        });
    }
}
```

c) Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Project Name: AdapterDemo

File Name: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">
   <Spinner
      android:id="@+id/spinner"
      android:layout_width="match_parent"
      android:layout_height="wrap_content" />
   <ListView
      android:id="@+id/listView"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:layout_marginTop="20dp" />
</LinearLayout>
```

File Name: MainActivity.java

```
package com.example.adapterdemo;
import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
```

```
import android.widget.ListView;
import android.widget.Spinner;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class MainActivity extends AppCompatActivity {
   Spinner spinner;
   ListView listView;
   String[] cities = {
         "Mumbai",
         "Pune",
         "Delhi",
         "Chennai",
         "Hyderabad"
   };
   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      spinner = findViewById(R.id.spinner);
      listView = findViewById(R.id.listView);

      // Spinner Adapter
      ArrayAdapter<String> spinnerAdapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_spinner_dropdown_item,
            cities
      );
      spinner.setAdapter(spinnerAdapter);
      // ListView Adapter
      ArrayAdapter<String> listAdapter = new ArrayAdapter<>(
```

```
        this,
        android.R.layout.simple_list_item_1,
        cities
    );
    listView.setAdapter(listAdapter);
    // Spinner Item Selection
    spinner.setOnItemSelectedListener(
        new AdapterView.OnItemSelectedListener() {
          @Override
          public void onItemSelected(
                AdapterView<?> parent,
                View view,
                int position,
                long id) {

             Toast.makeText(
                   MainActivity.this,
                   "Selected: " + cities[position],
                   Toast.LENGTH_SHORT
             ).show();
          }

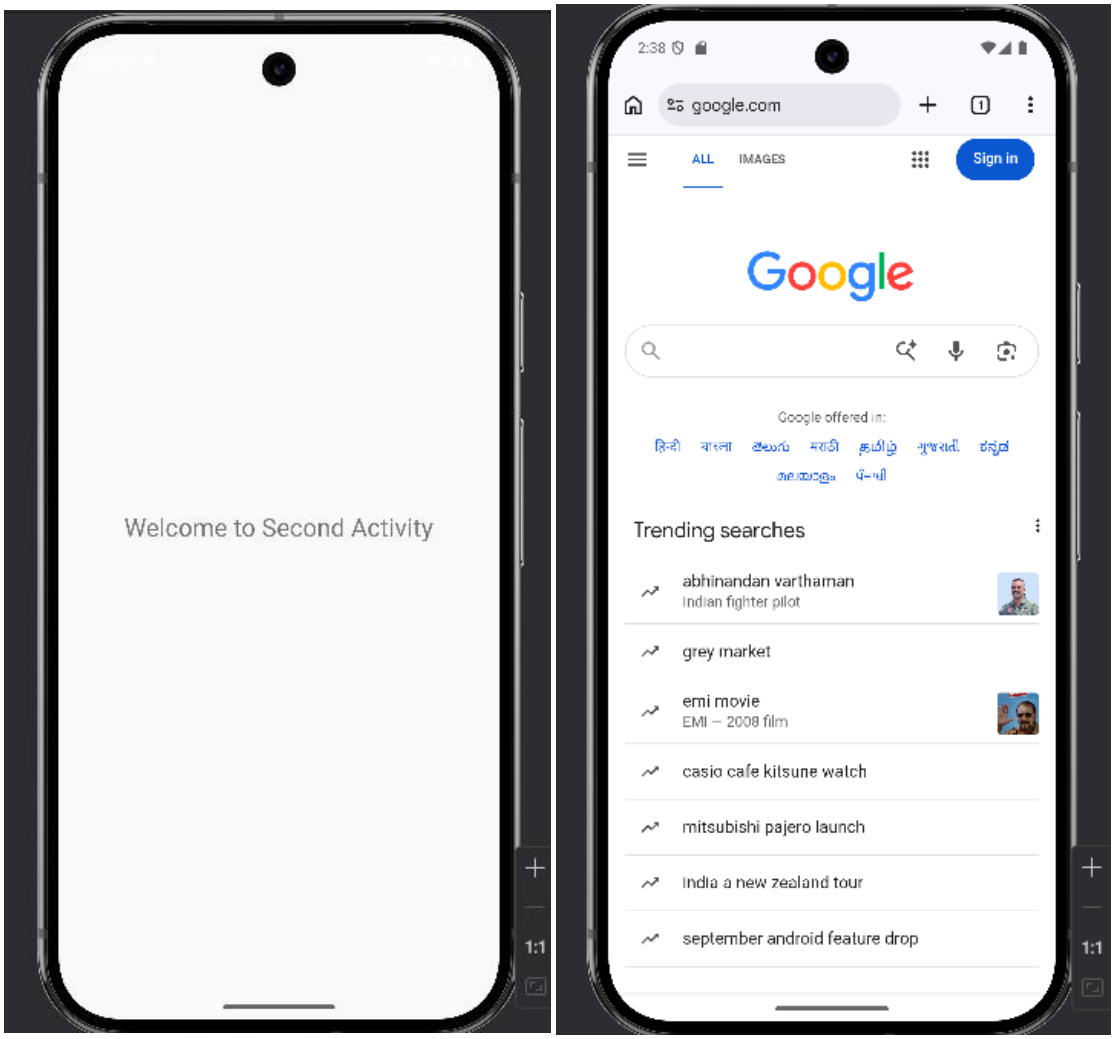
          @Override
          public void onNothingSelected(
                AdapterView<?> parent) {
          }
        });

    // ListView Item Click
    listView.setOnItemClickListener(
        (parent, view, position, id) -> {
```

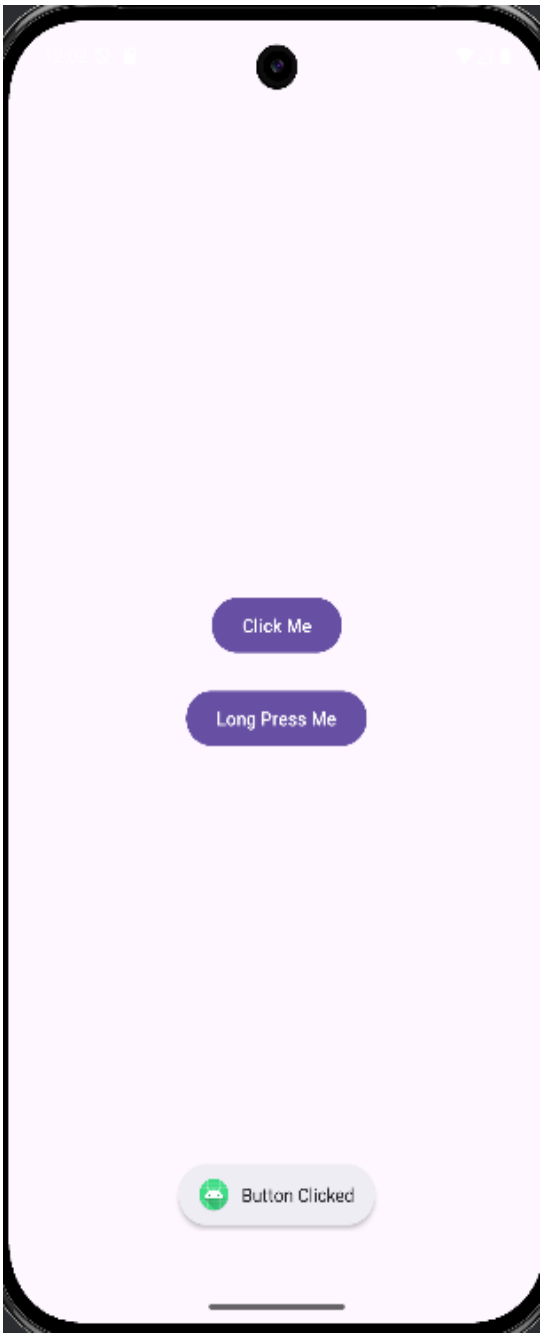
```
            Toast.makeText(
                    MainActivity.this,
                    "Clicked: " + cities[position],
                    Toast.LENGTH_SHORT
            ).show();
        });
    }
}
```

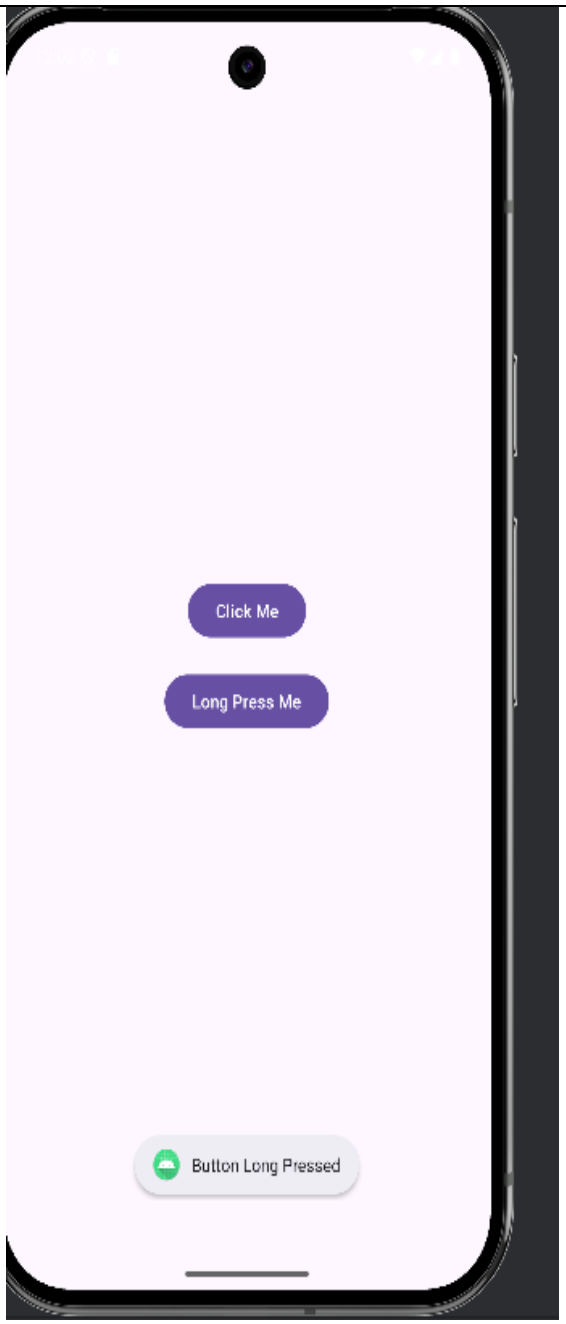
A) Output :




Welcome to Second Activity
2:38
google.com
ALL
IMAGES
Sign in
Google
Google offered in:
Trending searches
abhinandan varthaman
Indian fighter pilot
grey market
emi movie
EMI – 2008 film
casio cafe kitsune watch
mitsubishi pajero launch
india a new zealand tour
september android feature drop
1:1
1:1

B) Output :


Click Me
Long Press Me
Button Clicked


Click Me
Long Press Me
Button Long Pressed

C) Output :

